Hints: Limits – 6. Asymptotics and sharp estimates: sums, integrals, Stirling

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Find a telescoping sequence that is very close to $\frac{1}{k^2}$. For example, $\frac{1}{k-\frac12}-\frac{1}{k+\frac12}=\frac{1}{k^2-\frac14}$.

Hint 2. So $\sum_{k>n}\frac{1}{k^2-1/4}=\frac{1}{n+1/2}=\frac1n-\frac{1}{2n^2}+O(n^{-3})$. The difference $\frac{1}{k^2-1/4}-\frac{1}{k^2}=O(k^{-4})$ sums to $O(n^{-3})$ over $k>n$. (Euler–Maclaurin gives the same: integral $\frac1n$ minus the boundary term $\frac{f(n)}{2}$.)

Hint 3. For cubes, $\int_n^\infty x^{-3}dx-\frac12 n^{-3}$ suggests $\sum_{k>n}\frac{1}{k^3}=\frac{1}{2n^2}-\frac{1}{2n^3}+O(n^{-4})$. Prove it with Euler–Maclaurin, or by comparing with the telescoping $\frac12\big(\frac{1}{(k-\frac12)^2}-\frac{1}{(k+\frac12)^2}\big)=\frac{k}{(k^2-\frac14)^2}$.

Problem 2

Hint 1. Write $(1+\frac1n)^n=\exp\big(n\ln(1+\frac1n)\big)$ and expand the logarithm to the term in n^{-2}.

Hint 2. $n\ln(1+\frac1n)=1-\frac{1}{2n}+\frac{1}{3n^2}+O(n^{-3})$. Exponentiating gives $(1+\frac1n)^n=e\big(1-\frac{1}{2n}+\frac{11}{24n^2}+O(n^{-3})\big)$. In particular $(1+\frac1n)^n=e-\frac{e}{2n}+O(n^{-2})$.

Hint 3. Hence $n\big(e-(1+\frac1n)^n\big)\to\frac{e}{2}$.

Problem 3

Hint 1. Apply Stirling's formula in the sharp form of the class work, $n!=\sqrt{2\pi n}(n/e)^n e^{\theta_n}$ with $\theta_n=\frac{1}{12n}+O(n^{-2})$, to $(2n)!$ and to $n!$.

Hint 2. The main factors give $\frac{\sqrt{4\pi n}\,(2n/e)^{2n}}{2\pi n\,(n/e)^{2n}}=\frac{4^n}{\sqrt{\pi n}}$. The correction is $e^{\theta_{2n}-2\theta_n}$ with $\theta_{2n}-2\theta_n=\frac{1}{24n}-\frac{1}{6n}+O(n^{-2})=-\frac{1}{8n}+O(n^{-2})$.

Hint 3. Finally $e^{-1/(8n)+O(n^{-2})}=1-\frac{1}{8n}+O(n^{-2})$.

Problem 4

Note. The prime number theorem in the form $\pi(x)\sim x/\ln x$ gives only $p_n\sim n\ln n$; with that precision the error $o(n\ln n)$ swamps $n\ln\ln n$, and "$p_n\sim n\ln n+n\ln\ln n$" says nothing more than $p_n\sim n\ln n$. The hints explain the second term in the meaningful form $p_n=n\ln n+n\ln\ln n+O(n)$, which needs the PNT with a relative error, $\pi(x)=\frac{x}{\ln x}\big(1+O(\frac{1}{\ln x})\big)$.

Hint 1. Put $x=p_n$ into the PNT: $\pi(p_n)=n$, so $n\sim p_n/\ln p_n$. Take logarithms: $\ln n=\ln p_n-\ln\ln p_n+o(1)$, so $\ln p_n\sim\ln n$ and $p_n\sim n\ln p_n\sim n\ln n$.

Hint 2. The correction comes from not replacing $\ln p_n$ by $\ln n$ too early. From $p_n=n\ln p_n\,(1+O(1/\ln n))$ (the stronger PNT) you get $\ln p_n=\ln n+\ln\ln p_n+O(1/\ln n)=\ln n+\ln\ln n+o(1)$.

Hint 3. Substitute back: $p_n=n(\ln n+\ln\ln n+o(1))(1+O(1/\ln n))=n\ln n+n\ln\ln n+O(n)$. So the term $n\ln\ln n$ comes from the factor $\ln p_n$ itself being $\ln n+\ln\ln n$.

Problem 5

Hint 1. Integrate by parts to get $a_n=\frac{n-1}{n}a_{n-2}$, with $a_0=\frac{\pi}{2}$ and $a_1=1$.

Hint 2. Then $na_n a_{n-1} = (n-1)a_{n-1}a_{n-2}$, so this quantity does not depend on n; it equals $1 \cdot a_1 a_0 = \frac{\pi}{2}$.

Hint 3. Since $0 \leqslant \sin x \leqslant 1$, (a_n) decreases: $a_{n+1} \leqslant a_n \leqslant a_{n-1}$. Multiply by a_n and use the identity to get $\frac{\pi}{2(n+1)} \leqslant a_n^2 \leqslant \frac{\pi}{2n}$. Hence $a_n \sim \sqrt{\frac{\pi}{2n}}$.

Problem 6

Hint 1. For the first part show that $H_n - \ln n$ is decreasing and positive (compare $\frac{1}{k}$ with $\int_k^{k+1} \frac{dx}{x}$). For the integral, write everything as integrals over $[0,1]$: $\frac{1}{k} = \int_0^1 x^{k-1}dx$, so $H_n = \int_0^1 \frac{1-x^n}{1-x}dx$.

Hint 2. Also $\ln(n+1) = \int_0^1 \frac{x^n-1}{\ln x}\,dx$: differentiate in n (both sides have derivative $\frac{1}{n+1}$) and note that both vanish at $n=0$.

Hint 3. So $H_n - \ln(n+1) = \int_0^1 (1-x^n)\,g(x)\,dx$ with $g(x) = \frac{1}{1-x} + \frac{1}{\ln x}$. Check that g is bounded on $(0,1)$ (it tends to 1 at 0 and to $\frac{1}{2}$ at 1). Then $\int_0^1 x^n g \to 0$, and $\ln(n+1) - \ln n \to 0$.

Problem 7

Hint 1. This is $n\big(\frac{1}{n}\sum f(\frac{k}{n}) - \int_0^1 f\big)$ for $f(x) = x^p$, so expect $\frac{f(1)-f(0)}{2} = \frac{1}{2}$. But x^p is not smooth at 0 when $p < 1$, so use a different route: Stolz's rule.

Hint 2. The expression equals $\frac{A_n}{n^p}$ with $A_n = \sum_{k\leqslant n} k^p - \frac{n^{p+1}}{p+1}$, and $n^p \uparrow \infty$. Apply Stolz: compute $A_{n+1} - A_n = (n+1)^p - \frac{(n+1)^{p+1}-n^{p+1}}{p+1}$ by the binomial expansion.

Hint 3. $A_{n+1} - A_n = \frac{p}{2}n^{p-1} + O(n^{p-2})$ and $(n+1)^p - n^p = pn^{p-1} + O(n^{p-2})$. The limit is $\frac{1}{2}$ for every $p > 0$.

Problem 8

Hint 1. Extract the terms one at a time, applying Stolz each time to the remainder divided by the next candidate term. First, $\frac{\ln n!}{n\ln n} \to 1$, since $\frac{\ln(n+1)}{(n+1)\ln(n+1) - n\ln n} \to 1$.

Hint 2. Next, $\frac{\ln n! - n\ln n}{n} \to -1$, because the difference of the numerator is $-n\ln(1+\frac{1}{n}) \to -1$. Then $\frac{\ln n! - n\ln n + n}{\ln n} \to \frac{1}{2}$, because the difference of the numerator is $1 - n\ln(1+\frac{1}{n}) = \frac{1}{2n} + O(n^{-2})$ and that of $\ln n$ is $\frac{1}{n} + O(n^{-2})$.

Hint 3. Stolz cannot produce the constant itself. For $d_n = \ln n! - n\ln n + n - \frac{1}{2}\ln n$ compute $d_{n+1} - d_n = 1 - (n+\frac{1}{2})\ln(1+\frac{1}{n}) = -\frac{1}{12n^2} + O(n^{-3})$. The increments are summable, so $d_n \to C$.

Problem 9

Hint 1. On this interval tan increases from $-\infty$ to $+\infty$, and $\tan x - x$ is increasing there (its derivative is $\tan^2 x \geqslant 0$), so the root is unique. For $n \geqslant 1$ it lies close to the right end. Put $q = \pi n + \frac{\pi}{2}$ and $x_n = q - \varepsilon_n$ with $0 < \varepsilon_n < \frac{\pi}{2}$.

Hint 2. Then $\tan x_n = \cot \varepsilon_n$, so the equation is $\cot \varepsilon_n = q - \varepsilon_n$. Hence $\varepsilon_n \to 0$ and $\varepsilon_n \sim \frac{1}{q}$. Iterate: $\tan \varepsilon_n = \frac{1}{q-\varepsilon_n}$ gives $\varepsilon_n = \frac{1}{q} + \frac{2}{3q^3} + O(q^{-5})$. (Use $\arctan u = u - \frac{u^3}{3} + \dots$ and $\frac{1}{q-\varepsilon} = \frac{1}{q} + \frac{\varepsilon}{q^2} + \dots$.)

Hint 3. Now expand in n: $\frac{1}{q} = \frac{1}{\pi n} - \frac{1}{2\pi n^2} + O(n^{-3})$. So $x_n = \pi n + \frac{\pi}{2} - \frac{1}{\pi n} + \frac{1}{2\pi n^2} + O(n^{-3})$.

Problem 10

Hint 1. Apply Euler–Maclaurin to $f(x) = x^{-1/2}$: $\sum_{k=1}^n f(k) = \int_1^n f + \frac{f(1)+f(n)}{2} + \frac{B_2}{2}\big(f'(n) - f'(1)\big) + R_n$, where the remainder involves $f''' \sim x^{-7/2}$. The first-step remainder with f'' also works if you only need $O(n^{-3/2})$.

Hint 2. The terms are $2\sqrt{n}-2$ and $\frac{1}{2\sqrt{n}}$; the remaining ones contribute a constant plus $O(n^{-3/2})$. So $\sum_{k\leqslant n}\frac{1}{\sqrt{k}} = 2\sqrt{n} + C + \frac{1}{2\sqrt{n}} + O(n^{-3/2})$. It remains to identify C.

Hint 3. Do the same for k^{-s} with real $s > 1$. Then $Z(s) = \lim_n\bigl(\sum_{k\leqslant n} k^{-s} - \frac{n^{1-s}}{1-s}\bigr)$ exists for all $s > 0$, $s \neq 1$, and equals $\zeta(s)$ for $s > 1$, because $n^{1-s} \to 0$ there. The Euler–Maclaurin remainder is an integral that converges locally uniformly, so Z is analytic on $\operatorname{Re} s > 0$, $s \neq 1$. Thus Z is the analytic continuation of ζ, and $C = Z(\frac{1}{2}) = \zeta(\frac{1}{2}) \approx -1.4604$. The series $\sum k^{-1/2}$ itself diverges; $\zeta(\frac{1}{2})$ is the constant term of its partial sums.